
KGGDM: KNOWLEDGE GRAPH GUIDED DIFFUSION MODELS

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ABSTRACT

Recent advances in generative diffusion models have demonstrated remarkable capabilities in high-quality image synthesis. However, their controllability—particularly over fine-grained semantic attributes—remains a challenge [1]. In this work, we propose Knowledge Graph-Guided Diffusion Models (KGGDM), a novel framework that integrates structured knowledge from a semantic ontology into the image generation process. Using the CelebA dataset as a testbed, we construct a knowledge graph from binary attribute annotations, learn entity embeddings via TransE, and compress per-image attribute vectors through an autoencoder. These compact semantic representations are then projected into the conditioning space of a pre-trained Stable Diffusion model. We fine-tune the model to generate images conditioned on these knowledge graph-derived vectors, enabling nuanced and interpretable control over generated content. Our results underscore the potential of structured knowledge representations in enhancing the semantic control of generative models.



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1 Introduction

Diffusion models have recently emerged as a powerful class of generative models, producing highly realistic images across a variety of domains. Despite these advancements, controlling the content of generated images especially with fine-grained semantic details remains a major challenge.

Most conditioning techniques rely on either raw attribute vectors or natural language prompts. While effective, these approaches often fail to capture deeper relationships between attributes or offer interpretable control. For example, the attributes *Blond Hair*, *Smiling*, and *Wearing Lipstick* may be correlated, but are treated independently in standard conditioning methods.

In this paper, we propose **KGGDM** (Knowledge Graph-Guided Diffusion Models), a framework that introduces structured semantic knowledge into the image generation process. We construct a knowledge graph from binary facial attributes in the CelebA dataset, organize them into a human-readable ontology, and embed them using the TransE algorithm. These embeddings are then compressed with an autoencoder and used to condition a Stable Diffusion model during training.

Our method enhances the model’s ability to understand and respond to complex combinations of attributes, enabling better controllability and interpretability. Through experiments, we demonstrate that KGGDM outperforms baseline models in terms of image quality and attribute alignment, highlighting the benefits of using structured knowledge in generative modeling. Our work builds upon the foundational ideas presented in A.Gupta KGGLDM: Knowledge Graph Guided Diffusion Models for Advanced Learning, who introduced the integration of knowledge graphs into latent diffusion models.

2 Background

Knowledge Graphs (KGs) and Diffusion Models (DMs) are two transformative AI paradigms, each excelling in distinct areas: KGs structure domain knowledge through entities and relationships, enabling semantic reasoning, while DMs generate high-fidelity data by iteratively denoising random noise. Despite their individual successes, KGs in NLP and recommendation systems, DMs in surpassing GANs / VAEs for image / audio synthesis, their integration remains underexplored. This Project investigates how KG semantics can enhance DMs, improving controllability and contextual alignment in generative tasks.

2.1 Knowledge Graphs (KGs)

A knowledge graph is a structured representation of knowledge that captures entities, their attributes, and the relationships between them in a graph-based format. In artificial intelligence (AI), knowledge graphs serve as a powerful framework for organizing and reasoning over complex information, enabling machines to interpret and retrieve data in a semantically meaningful way.

Typically, a knowledge graph consists of nodes (representing entities, concepts, or events in the real world) and edges (denoting the relationships between them). This allows efficient data integration, semantic search, and inference, making it valuable in applications such as natural language processing, recommendation systems, and Generative AI. By encoding domain-specific or general-world knowledge in a machine-readable format. Knowledge graphs improve the AI system’s ability to understand context, reduce ambiguity, and support decision-making processes.

The flexibility and scalability of knowledge graphs make them a key component in modern AI systems for representing and reasoning over interconnected data.

2.2 Knowledge Graph Embeddings

Knowledge graph embeddings are dense, low-dimensional vector representations that encode entities and relations from a knowledge graph into a continuous vector space. Unlike symbolic representations that treat entities and relationships as discrete symbols, embeddings capture their semantic properties through numerical vectors. The primary objective of knowledge graph embeddings is to preserve the structural and relational patterns of the graph, such as hierarchy, symmetry, and compositionality, while facilitating tasks such as link prediction, entity resolution, and knowledge graph completion.

Several embedding techniques have been developed to project knowledge graphs into vector spaces, including translational models (e.g., TransE, TransR), tensor factorization approaches (e.g., RESCAL, DistMult), and neural methods (e.g., Graph Neural Networks, Transformer-based embeddings). By transforming discrete relational data into continuous

representations, they bridge the gap between symbolic knowledge and statistical machine learning, enhancing both interpretability and predictive performance.

2.3 Diffusion Models (DMs)

Diffusion models are a class of generative machine learning models that learn to process data by denoising random noise over multiple steps. These models simulate a two-phase process: a forward diffusion process, where input data is incrementally corrupted with noise, and a reverse diffusion process, where the model learns to remove noise to reconstruct the original data distribution.

In the forward phase, the model applies a fixed or learned noise schedule to transform structured data (such as images or text) into random Gaussian noise over several timesteps. The reverse phase is typically implemented using deep neural networks and is trained to predict and reverse this corruption, effectively generating new samples by refining noise into coherent output iteratively. Due to their iterative refinement approach, diffusion models excel at producing high-quality, diverse samples, making them effective in image generation.

2.4 Knowledge Graph guided Diffusion Models

Recent advances in generative AI have introduced some new methods to improve the precision and reliability of diffusion models by including structured knowledge representations. Among these, knowledge-graph-guided diffusion models have emerged as a powerful framework that combines the iterative refinement capabilities of diffusion processes with the semantic richness of knowledge graphs. At their core, knowledge graph-guided diffusion models operate by conditioning the denoising process on structured relational data. Unlike traditional diffusion models that learn purely from unstructured data distributions, these models integrate explicit knowledge representations.

For the context of this project, in image generation, spatial and compositional relationships extracted from knowledge graphs ensure that generated scenes adhere to real-world physics and semantics. The guidance mechanism typically involves mapping knowledge graph embeddings into the diffusion model’s latent space, either through cross-attention layers or encoders. This allows the model to dynamically adjust the noise prediction based on the relational context. In particular, this not only improves output quality but also enhances interpretability.

3 Objectives

- **Dataset Preparation** : Curate and preprocess the CelebA dataset, ensuring high-quality facial images and accurate extraction of metadata for ontology development.
- **Ontology and Knowledge Graph (KG) Construction** : Design a hierarchical ontology of facial attributes (e.g., hair color, accessories) and construct a KG capturing relationships among these attributes using CelebA metadata.
- **Embedding Generation and Feature Compaction** : Generate dense vector representations of KG entities and relationships using the TransE model, followed by dimensionality reduction via an autoencoder to produce compact, semantically rich conditioning signals.
- **Integration with Diffusion Models** : Incorporate the compacted KG embeddings into the Stable Diffusion framework to guide attribute-aware image generation, enhancing fidelity and controllability.
- **Evaluation and Validation** : Assess model performance using metrics like Frechet Inception Distance (FID) and Inception Score (IS), alongside conditional metrics to verify attribute adherence. Conduct ablation studies to quantify the impact of KG guidance.

4 Methodology

In this section, we present our full pipeline, **KGGDM (Knowledge Graph-Guided Diffusion Models)**, for semantically controllable image generation. Our approach integrates structured semantic knowledge, encoded as a knowledge graph, into the conditioning mechanism of a pre-trained diffusion model. The pipeline consists of five key stages: dataset preparation, knowledge graph construction and embedding, attribute vector compaction, Stable Diffusion modification, and fine-tuning. The use of knowledge graphs to enhance text-to-image diffusion models has been shown to improve context-awareness and semantic consistency in generated images, further supporting the effectiveness of our approach [3].

4.1 Dataset Preparation

The CelebA dataset comprises over 200,000 facial images, each annotated with 40 binary attributes (e.g. *Smiling*, *Blond Hair*, *Eyeglasses*). We center-crop and resize each image to 256×256 pixels, then normalize pixel values to $[-1, 1]$. Attribute labels are loaded into a pandas `DataFrame` for fast querying during graph construction and training.

To investigate gender bias, we split CelebA into “male” and “female” subsets and computed per-attribute prevalences. Figure 1 shows:

- The ten attributes most over-represented in female images;
- The ten attributes most over-represented in male images;
- The raw difference (female minus male prevalence) across all 40 attributes;
- A global side-by-side comparison of male versus female prevalence for all 40 attributes.

We see extreme skews on *Wearing Lipstick* and *Heavy Makeup* (80 % of females vs. < 1 % of males), and the converse on *5_o_Clock_Shadow*, with many more moderate gaps elsewhere. These pervasive imbalances motivate fairness-aware modeling.

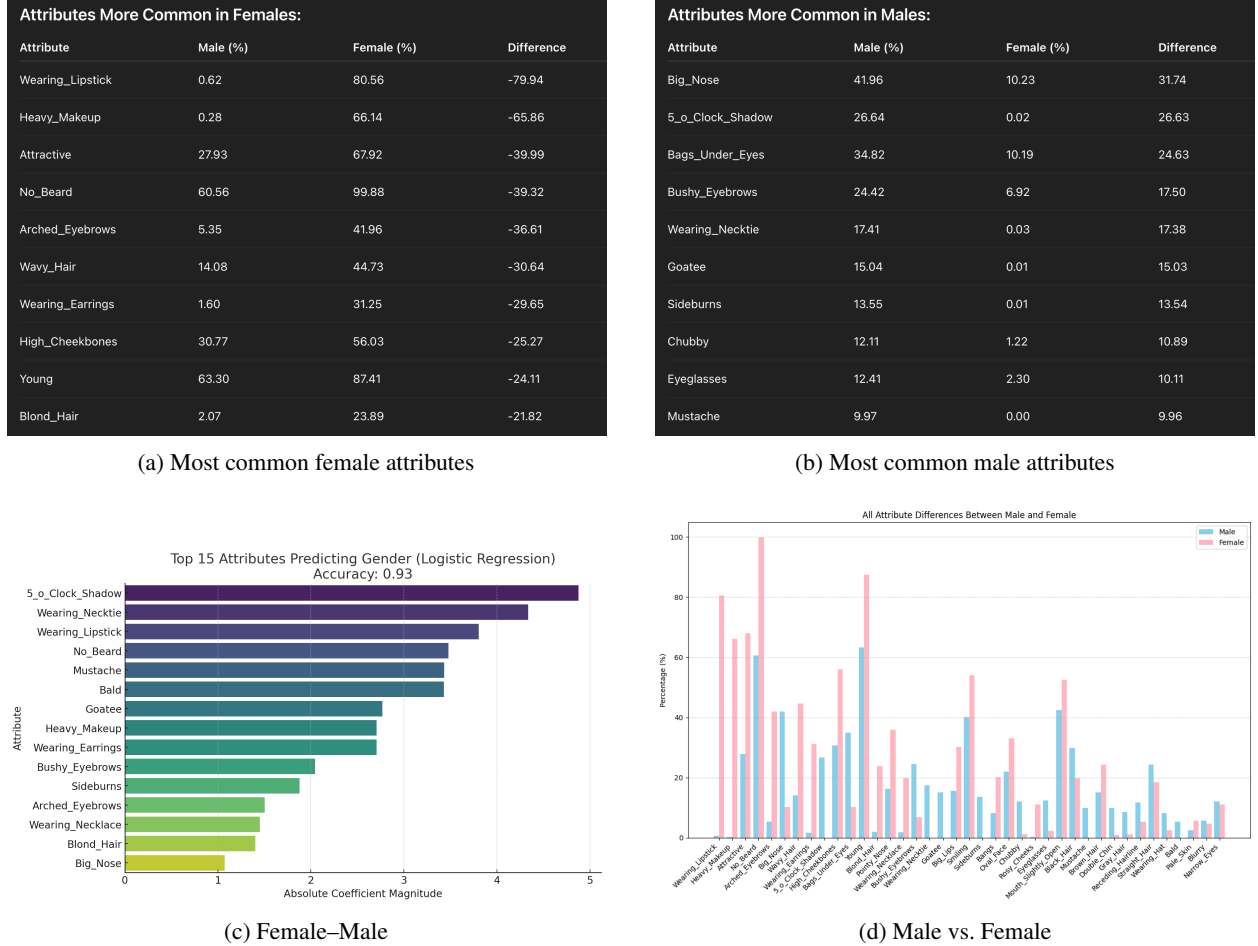


Figure 1: Attribute distribution disaggregated by gender. (a) Top-10 attributes skewed to females; (b) top-10 skewed to males; (c) raw difference (female minus male); (d) side-by-side prevalences for all 40 attributes.

4.2 Knowledge Graph Construction

To encode structured semantic relationships, we construct a knowledge graph (KG) using the CelebA attribute labels. Each image is treated as an entity, and its positive attributes (+1) are connected via typed relations:

```
(image_ID, has_attribute, attribute_name)
```

The KG is represented in a triple format, which consist a subject, a predicate and object as shown in brackets above. Additionally, attributes are grouped into semantic categories (e.g., Hair, Makeup, Accessories), allowing us to define higher-level ontology relations between attributes themselves. (see Fig 2).

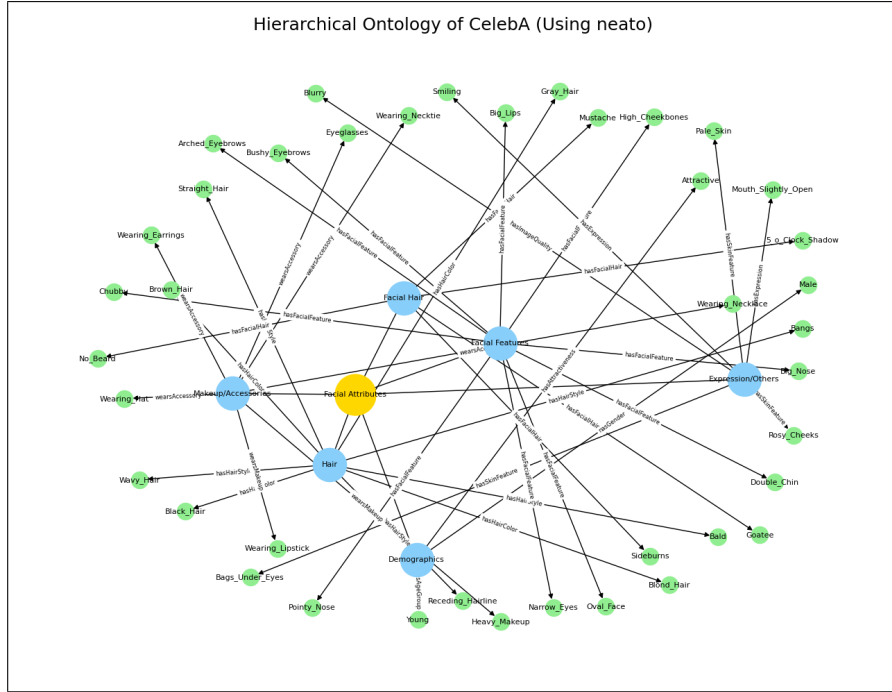


Figure 2: Ontology

After the construction of KG ontology, our next task is to create KG embeddings which is important for the semantics of the vector space. KG embeddings help in integration of the KG with machine learning models and also helps in computing similarities. We used PyKEEN for this process.

Before PyKEEN can create embeddings, we have to set up our knowledge graph data properly. In simple terms, the data must be a bunch of simple "A relates to B" statements, organized in a spreadsheet-like file (CSV or TSV). So, we have to take those statements we built before and save them into one of these files, double-checking that all the names (for the "A"s, "B"s, and the relationships) are saved accurately. We exported the triples we made in the last step into CSV format, checking that all the entities and relationships are written correctly.

4.3 Knowledge Graph Embedding (TransE)

To capture how different attributes are related to each other, we used a method called TransE (Translating Embeddings) for knowledge graph embedding. In simple terms, TransE turns all the attributes and their relationships into vectors—lists of numbers—so that the computer can understand and work with them in a continuous space.

TransE model operates on the principle that for a triplet (h,r,t), where h is the head entity, r is the relation, and t is the tail entity, the embedding should satisfy :

$$h + r \approx t$$

Negative Sampling : For each valid triplet (h, r, t) , corrupt the tail entity to generate negative samples (h, r, t') , where t' is randomly selected.

Loss Function: Minimize the margin-based ranking loss:

$$C = \sum_{(h,r,t)} \max(0, \gamma + ||h + r - t|| - ||h + r - t'||)$$

where γ is a hyperparameter defining the margin between positive and negative triplets.

Embedding Dimension: The dimensionality d of embeddings is tuned to balance expressiveness and computational efficiency ($d=100$).

Optimization: Adam optimizer is used with a learning rate of 0.001 and early stopping based on validation triplet accuracy.

We trained the embeddings using PyKEEN. During training, the model learned to assign vectors so that real (correct) combinations of attributes and relations ended up close together, while incorrect combinations were farther apart. This was achieved by minimizing a loss function that compares correct and randomly corrupted triples. The objective function minimized a margin-based ranking loss, encouraging correct triples to have lower energy (L2 distance) compared to corrupted triples where either the head or tail was randomly replaced. This formulation allowed the model to learn embeddings that preserved the relational semantics of attributes in a latent space of dimension 50.

After training, the resulting attribute embeddings were exported and stored as a matrix of shape 40 x 50. These embeddings became a key input to our autoencoder and image generation pipeline, providing a way to condition the image generation process on specific semantic concepts. This helped the system generate images that better reflected combinations of chosen attributes. Fig 3 shows the likliness of similarities of (Male, ?, Attractive) to different attributes that can replace the '?'.²

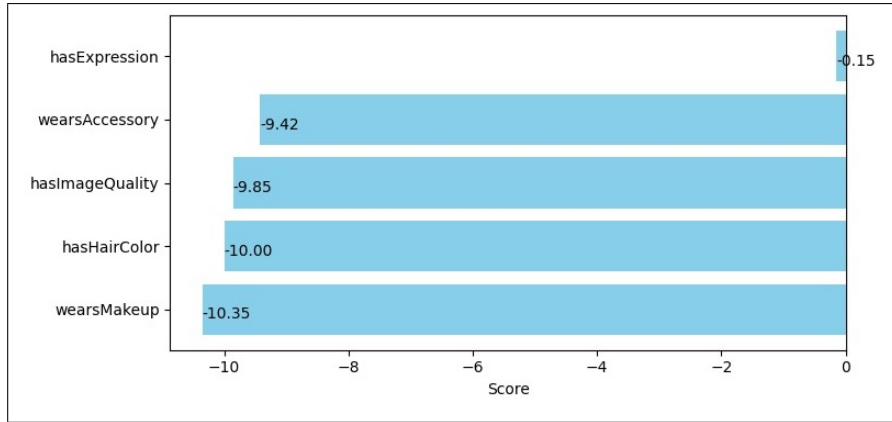


Figure 3: Predictions for (Male, ?, Attractive). Shows which attributes are not closely related to the triple (negative score)

4.4 Attribute Vector Compaction with Autoencoder

To create a compact and expressive representation of attribute combinations, we implemented an autoencoder that compresses high-dimensional semantic vectors into a low-dimensional latent space. This transformation enables efficient conditioning of the diffusion model while preserving relevant attribute information. To obtain a fixed-size conditioning vector for each image:

- We construct a 40×50 attribute matrix for each image by multiplying the embedding of each attribute by +1 or -1 based on its binary label.
- This matrix is flattened into a high-dimensional vector of length 2000, which is then passed through an autoencoder.
- The encoder compresses the vector into a latent representation (128 dimensions), capturing essential semantic features.

During training, we minimized the mean squared error (MSE) between the reconstructed semantic component and the original semantic part of the input vector. Training proceeded over 50 epochs using the Adam optimizer with a learning rate schedule that reduced the learning rate upon plateauing validation loss.

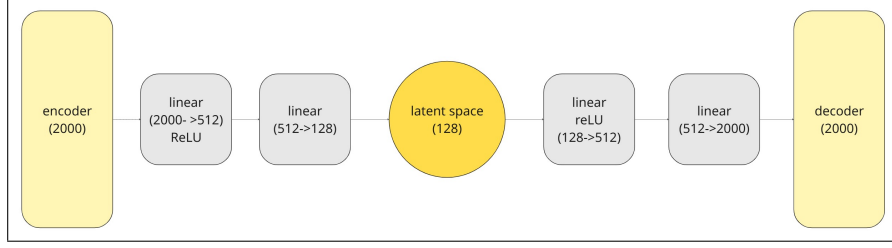


Figure 4: Autoencoder architecture

Once trained, the encoder was used to transform any target combination of attributes into a 128-dimensional compressed vector. This vector served as the conditioning input to the UNet diffusion model during image generation, enabling the model to synthesize images reflecting the desired semantic properties. This approach allowed the model to leverage both structured semantic attribute information and additional variation introduced by the random noise, resulting in more diverse and attribute-consistent generation.

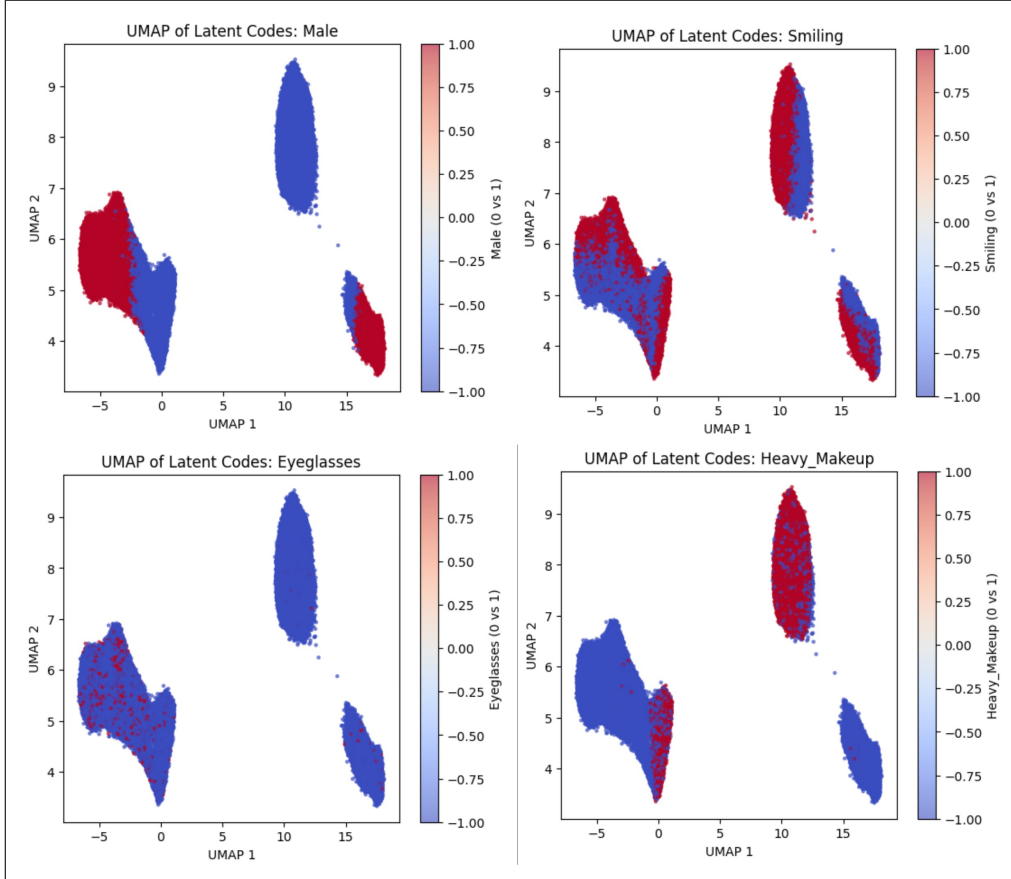


Figure 5: UMAP visualization of latent codes colored by binary attributes: *Male*, *Smiling*, *Eyeglasses* and *Heavy Makeup*. Distinct color regions indicate that attribute semantics are preserved in the compressed latent space.

4.5 Latent Space Visualization with UMAP

To interpret the structure of the compressed semantic space, we apply **Uniform Manifold Approximation and Projection (UMAP)** for dimensionality reduction. UMAP is a non-linear technique that preserves both local and global structure of high-dimensional data when projected into 2D space.

We extract latent codes from the trained autoencoder for the entire CelebA dataset. Let $X_{\text{aug}} \in \mathbb{R}^{N \times D}$ denote the attribute vectors after augmentation. We pass these vectors through the encoder, obtaining $Z \in \mathbb{R}^{N \times d_{\text{latent}}}$, where $d_{\text{latent}} = 128$ in our implementation.

The latent codes are then projected into a 2D space using UMAP:

$$Z_{\text{UMAP}} = \text{UMAP}(Z)$$

We use the following UMAP configuration: 30 neighbors, minimum distance of 0.1, and a fixed random seed for reproducibility.

The resulting UMAP embeddings are colored by different binary attributes (e.g., *Male*, *Smiling*, *Eyeglasses*) to assess clustering. As shown in Figure 5, certain attributes such as *Male* exhibit well-separated clusters, indicating that the latent space encodes attribute semantics effectively.

5 KGGDM Process

5.1 Stable Diffusion Modification

Our goal is to train the diffusion model to “listen” to a 128-dimensional semantic code, compressed via the autoencoder (Sec. 4.4), and guide image synthesis—while keeping the original generative prior largely intact.

To achieve this, we use the Stable Diffusion v1-5 architecture from HuggingFace’s `diffusers` library[7], which consists of three core components:

- **AutoencoderKL (VAE)** – maps images to and from a compressed latent space,
- **UNet2DConditionModel (U-Net)** – performs iterative denoising in latent space,
- **DDPMScheduler** – defines the forward/reverse noise schedule.

We freeze the VAE to preserve the pretrained image representations and fine-tune only the U-Net. To inject semantics into the generation process, we introduce a lightweight projection head—`Linear(128 → 768)`—that maps each semantic code into the U-Net’s cross-attention space. This allows us to condition the denoising process without architectural modifications.

5.2 Latent Diffusion Fine-Tuning

Once the semantic projection head is integrated, we proceed to fine-tune the U-Net on latent-space noise prediction, conditioned on knowledge-graph-derived semantic codes.

We precompute VAE latents for all CelebA images using the frozen encoder, and pair them with their respective 128-dimensional compressed attribute codes from the autoencoder. During training, each latent $\ell \in \mathbb{R}^{4 \times 64 \times 64}$ is corrupted with Gaussian noise at a randomly sampled timestep $t \sim \mathcal{U}\{0, T-1\}$, yielding a noised latent $\tilde{\ell}_t$. The model then attempts to reconstruct the noise vector ε from the noised latent and the projected semantic code.

The optimization objective is standard Denoising Score Matching, expressed as a mean squared error (MSE) between the predicted and true noise:

$$\mathcal{L}_{\text{MSE}} = \mathbb{E} \left[\|\varepsilon - \hat{\varepsilon}\|_2^2 \right], \quad \text{where} \quad \hat{\varepsilon} = \text{U-Net}_{\theta}(\tilde{\ell}_t, t, h)$$

Here, $\tilde{\ell}_t$ is the noised latent produced by the scheduler at timestep t , ε is the injected Gaussian noise, and h is the projected semantic conditioning vector obtained via a `Linear(128 → 768)` layer. The expectation is taken over image latents, noise samples, timesteps, and semantic codes.

We train the model for 5–10 epochs using the Adam optimizer with a learning rate of 1×10^{-4} . The U-Net and projection head are updated jointly, while the VAE remains frozen throughout. To improve training stability, we clip gradients to a maximum norm of 1.0 and periodically save checkpoints containing the U-Net, projection head, and optimizer state.

To accelerate training and reduce memory load, all VAE latents are cached in advance.

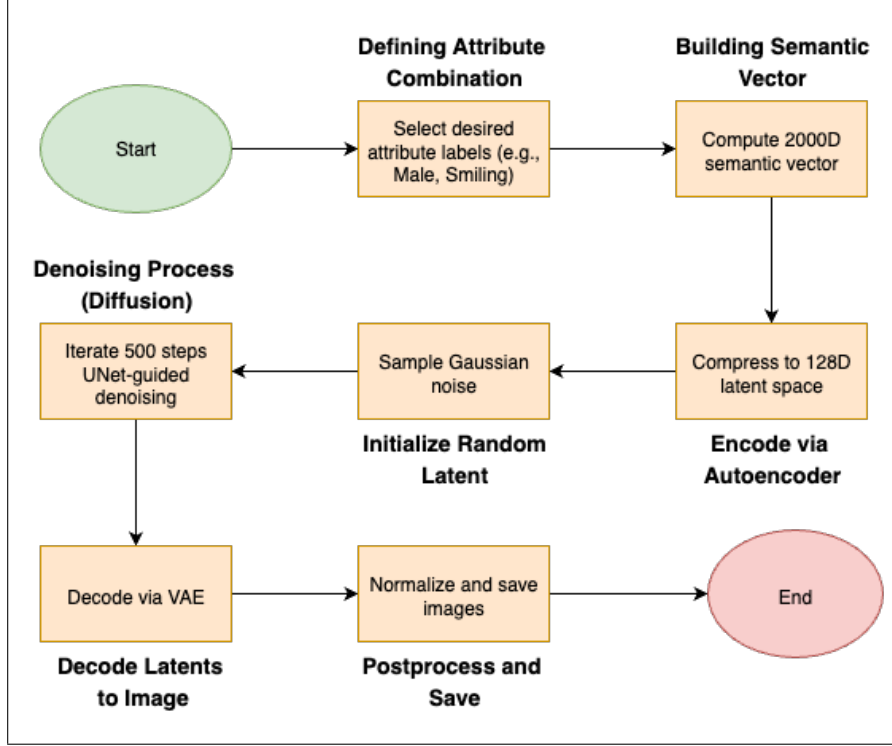


Figure 6: Image Generation flowchart

5.3 Image Generation

After completing the fine-tuning of the Latent Diffusion Model, we implemented an image generation pipeline to generate images conditioned on specific attribute combinations. The process uses the learned semantic embedding space and the projection mechanism integrated during training. For each target attribute configuration, a 40-dimensional binary label vector is constructed, where each dimension corresponds to one CelebA attribute. Each attribute vector indicates which facial characteristics are active in the prompt. For example, to generate images of young faces wearing lipstick with pale skin, the attributes “Young,” “Wearing_Lipstick,” and “Pale_Skin” were activated, and all others were set to zero.

These binary vectors were transformed into 2000-dimensional semantic vectors by element-wise multiplication with the precomputed attribute embedding matrix (dimension 40×50), followed by flattening. This semantic representation aims to capture relationships and similarities between attributes as learned from the knowledge graph embeddings. Afterwards, the semantic vectors were encoded into compressed 128-dimensional latent codes using the encoder module of the trained autoencoder. It outputs a 128-dimensional compressed representation, which serves as the conditioning input for image generation. No random noise component is concatenated at this stage to ensure deterministic encoding of the specified attributes.

The compressed embedding is repeated to form a batch of conditioning vectors of shape $(B, 128)$, where $B=20$ is the number of images per combination. These embeddings are projected through a learned linear projection and reshaped to match the UNet cross-attention input dimension. For each timestep of the diffusion process (500 inference steps), the model iteratively denoised the latents. In each iteration, the conditioning embeddings were projected into the cross-attention space using the trained projection head, enabling the UNet to guide the denoising process according to the specified attributes. After all timesteps were completed, the resulting latent representations were decoded back into RGB images using the pretrained Variational Autoencoder (VAE) decoder. Finally, the images were scaled to the $[0, 1]$ range and are saved individually as PNG files in subfolders named according to their active attribute combination. (Fig 7 shows all the steps in the process).

5.4 Evaluation Metrics

To evaluate the performance of our Knowledge Graph-Guided Diffusion Model (KGGDM), we employed both quantitative and qualitative assessment methods:

Fréchet Inception Distance (FID). Fréchet Inception Distance (FID) quantifies the difference between distributions of real and generated images

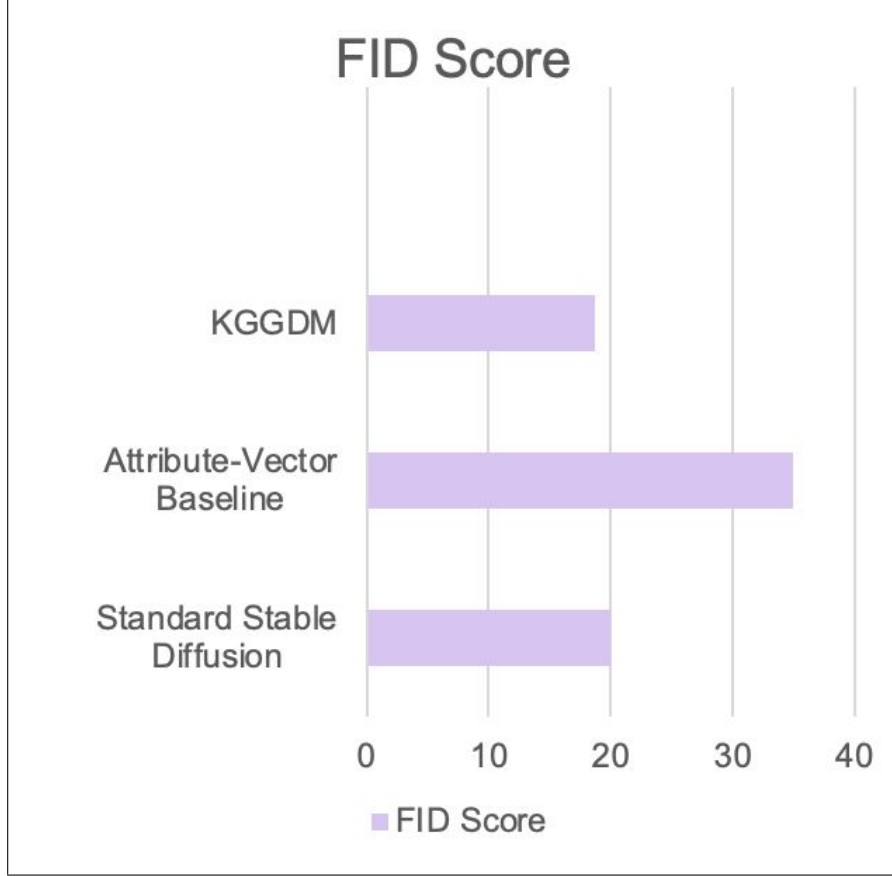


Figure 7: fid

Our model achieves an FID score of 18.746, compared to 35.0 for the baseline conditioned on raw attribute vectors. This demonstrates that incorporating structured semantic conditioning through knowledge graph embeddings substantially improves generative fidelity, yielding distributions of generated images that are markedly closer to real data.

Human Evaluation. To assess the effectiveness of our conditioned image generation pipeline, we conducted a series of evaluations across 6 distinct attribute combinations. We generated batches of 6 synthetic images per combination and assessed their semantic consistency with the target attributes. For interpretability, we report an approximate semantic similarity score, reflecting the proportion of generated samples that visibly align with the intended attributes. This evaluation provides both qualitative and quantitative insights into the model’s ability to control semantic content during generation. Each subsection below presents:

- Target Attribute Combination
- Sample Outputs
- Estimated Semantic Similarity
- Observations and Remarks

This structured approach enables transparent comparison of controllability across varying levels of attribute complexity.

Combination 1 : Female, Wearing_Lipstick
Sample Outputs: Fig 7



Figure 8: Images from Combination 1

Observations : Most generated images (**83.6%**) accurately depict Female subjects with visible Lipstick . A small subset shows ambiguous characteristics.

Combination 2 : Female, Wearing_Heavy_Makeup
Sample Outputs: Fig 8



Figure 9: Images from Combination 2

Observations : Most generated images (**83.6%**) accurately depict Female subjects with visible Makeup . A small subset shows ambiguous characteristics.

Combination 3 : Male, Bald
Sample Outputs: Fig 9



Figure 10: Images from Combination 3

Observations : Most generated images (**50.0%**) accurately depict Male subjects with visible Baldness. A small subset shows ambiguous characteristics.

Combination 4 : Male, Eyeglasses
Sample Outputs: Fig 10

Observations : Most generated images (**50%**) accurately depict Male subjects with Eyeglasses. A small subset shows ambiguous characteristics.



Figure 11: Images from Combination 4

Combination 5 : Female, Young, Pale_Skin
 Sample Outputs: Fig 11

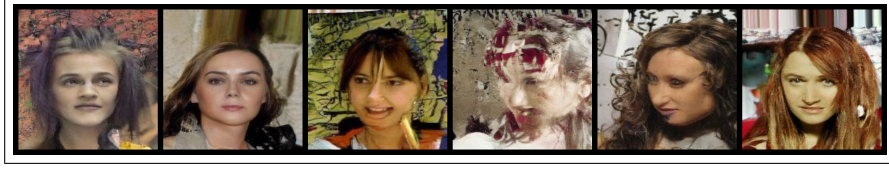


Figure 12: Images from Combination 5

Observations : Most generated images (**83.3%**) accurately depict Young Female subjects with Pale skin. A small subset shows ambiguous characteristics.

Combination 6 : Male, Wearing_Earrings, Wearing_Makeup, Rosy_Cheeks
 Sample Outputs: Fig 12



Figure 13: Images from Combination 6

Observations :Lots of combinations showed low semantic consistency and visual coherence. It tried to generate some random images with attributes that do not have a relation between them.

6 Discussion and Potential Problems

6.1 General Findings

The proposed KGGDM framework demonstrates the feasibility and promise of integrating structured semantic knowledge into diffusion-based generative models. By using knowledge graphs and embedding techniques, the model enables more interpretable and controllable image synthesis, moving beyond standard attribute or prompt-based conditioning.

- **Semantic Control:** The use of knowledge graph-derived embeddings as conditioning vectors allows the model to generate images that better reflect complex attribute combinations, as observed in qualitative results and preliminary metric comparisons.
- **Interpretability:** The structured approach provides more transparent and explainable conditioning, making it easier to understand how specific attributes or relationships influence the generated outputs.
- **Attribute Alignment:** Initial results suggest that KGGDM improves alignment between intended and generated attributes, especially for combinations that are semantically related but not explicitly encoded in raw attribute vectors.

6.2 Limitations

- **Limited Training Resources:** The model was trained for only 10 epochs due to computational constraints and the scope of a university experiment. This likely limited the convergence and overall performance, especially compared to state-of-the-art diffusion models trained on larger scales.
- **Hardware Constraints:** Memory and GPU limitations restricted batch sizes, model complexity, and the number of training iterations, potentially impacting the diversity and quality of generated images.
- **Dataset Bias:** The CelebA dataset exhibits strong attribute imbalances (e.g., gender, makeup, hair color), which may introduce bias into both the knowledge graph and the generated images. This could affect the fairness and generalizability of the approach.
- **Evaluation Scope:** Quantitative evaluation is currently limited and based on preliminary metrics. More comprehensive studies, including ablation experiments and user studies, are needed to fully validate the benefits of knowledge graph guidance.

6.3 Potential Improvements and Future Directions

- **Longer Training and Hyperparameter Tuning:** Extending training beyond 10 epochs, experimenting with different learning rates, batch sizes, and optimizer settings could yield significant improvements in image quality and attribute fidelity.
- **Larger or More Diverse Datasets:** Applying the method to larger or more diverse datasets would help assess generalization and robustness, as well as mitigate dataset-specific biases.
- **Advanced Knowledge Graph Construction:** Incorporating richer ontologies, external knowledge bases, or dynamic graph updates could further enhance semantic control and interpretability.
- **Model Architecture Enhancements:** Exploring alternative conditioning mechanisms (e.g., attention-based fusion, graph neural networks) or deeper autoencoder architectures may improve the integration of semantic information.
- **Comprehensive Evaluation:** Implementing a broader suite of quantitative metrics (e.g., FID, IS, CLIPScore) and conducting extensive human evaluations will provide a more rigorous assessment of model performance.
- **Fairness and Bias Mitigation:** Explicitly addressing dataset biases through balanced sampling, fairness-aware loss functions, or post-hoc corrections could improve the ethical aspects of generated content.

6.4 Conclusion

Despite resource constraints and the experimental nature of this study, the KGGDM framework highlights the value of structured semantic knowledge in guiding diffusion-based generative models. Further research, scaling, and evaluation are warranted to fully realize its potential and address the identified limitations.

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